

Marking Scheme — General Science Sample Test

Maximum marks: 20 · Award marks for correct points even if worded differently.

Q1. [2 marks]

- Light bends / changes direction (1)
- when it passes from one transparent medium to another, because its speed changes (1)

Q2. [3 marks]

- Formula: speed = distance ÷ time (1)
- Substitution: $120 \div 2$ (1)
- Answer: 60 km/h, with the unit (1)

Q3. [4 marks]

- First law: a body stays at rest or in uniform motion unless an external force acts (inertia) (1)
- Second law: $F = ma$, or force = rate of change of momentum (1)
- Third law: every action has an equal and opposite reaction (1)
- A valid example of the third law, e.g. recoil of a gun, rowing a boat, rocket launch, walking (1)

Q4. [5 marks]

- Cell drawn with correct symbol (1)
- Switch drawn with correct symbol (1)
- Bulb drawn with correct symbol (1)
- All three labelled and joined in series in one closed loop (1)
- Explanation: the circuit is complete, current flows and the bulb glows (1)

Q5. [6 marks]

- Definition: green plants make their food (glucose) from carbon dioxide and water using sunlight (2; give 1 if only part of this)
- Equation: correct reactants and products (1)
- Equation balanced: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (1)
- Chlorophyll absorbs light energy (1)
- Chlorophyll is found in chloroplasts / gives leaves their green colour (1)

Suggested checking options: Step marks · Marks for diagram alone · Ignore spelling & grammar.